



LABORATORY REPORT



Contract Number: PSL26/3342

Report Date: 20 April 2026

Client's Reference:

Client Name: Sirius Leeds
4245 Park Approach
Century Way
Thorpe Park
Leeds
LS15 8GB

For the attention of: Ami Cooper

Contract Title: Land off Tinker Lane, Rufforth, York, YO23 3QA

Date Received: 17/4/2026

Date Commenced: 17/4/2026

Date Completed: 20/4/2026

Notes: Opinions and Interpretations are outside the UKAS Accreditation

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Checked and Approved Signatories:

A Watkins
(Managing Director)

R Berriman
(Associate Director)

S Royle
(Laboratory Manager)

L Knight
(Assistant Laboratory Manager)

S Eyre
(Senior Technician)

T Watkins
(Senior Technician)

5 – 7 Hexthorpe Road,
Hexthorpe,
Doncaster,
DN4 0AR
Tel: 01302 768098
Email: rberriman@prosoils.co.uk
awatkins@prosoils.co.uk

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VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test: 17-Apr-26

Test Ref: PBT 1

Grid Ref:

Depth (m): 0.35

Layer:

Conditions: Dry

Maximum Applied Pressure (kPa):

205.85

Maximum Deformation (mm):

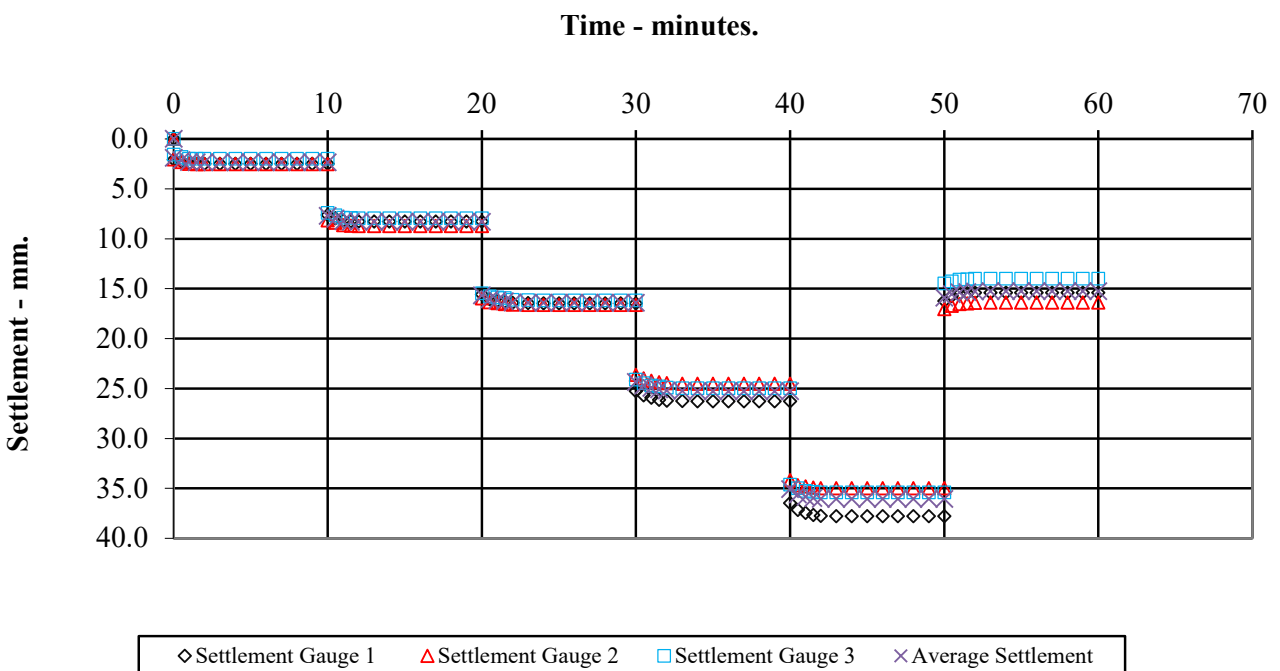
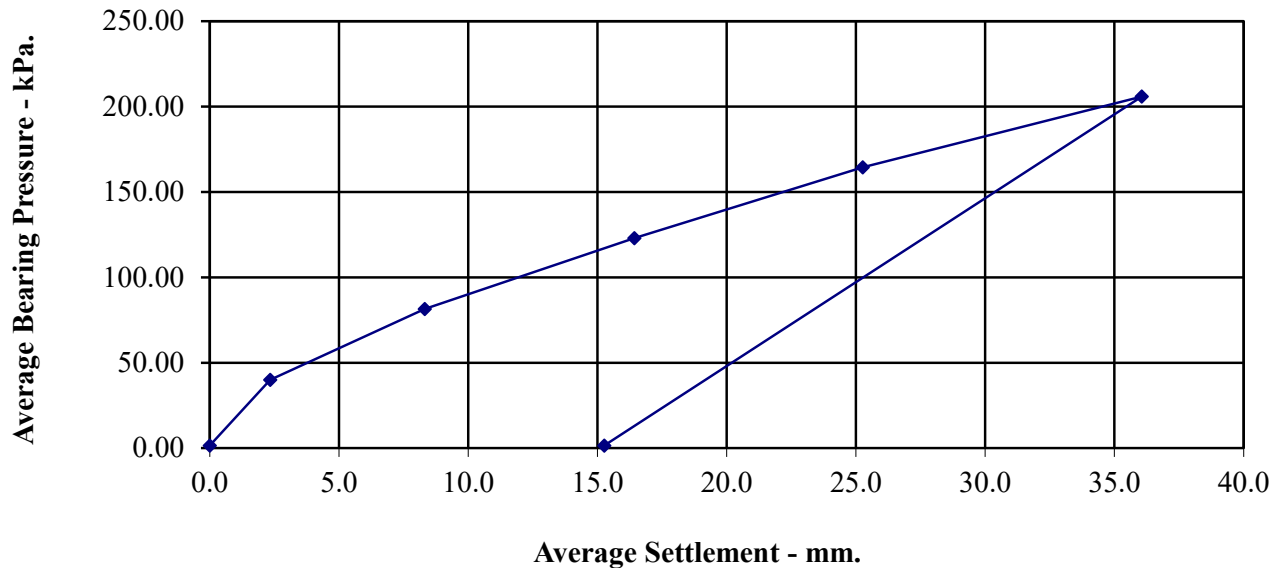
36.06

Plate Area (m²):

0.2922

Description:

Brown CLAY



Calculation of Equivalent CBR Value from Plate Bearing Test

Design Manual for Roads and Bridges Volume 7: Pavement Design and Maintenance.

HD 25/94: Foundations

Date of Test 17-Apr-26

Test Ref PBT 1

Depth (m) 0.35

Grid Ref

Layer

Conditions Dry

Description Brown CLAY

Maximum Deflection 36.06 mm

Deflection required for CBR value 1.25 mm

Load at 1.25mm 22 kN/m²

Plate diameter 610 mm

Conversion factor for plate diameter 0.816

**K₇₆₂(modulus of subgrade reaction)
calculated using 1.25mm settlement** 14.4 kN/m²/mm

CBR Value 1.0 %



Land off Tinker Lane, Rufforth, York

Contract No:

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Client Ref:

VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test: 17-Apr-26

Test Ref: PBT 2

Grid Ref:

Depth (m): 0.45

Layer:

Conditions: Dry

Maximum Applied Pressure (kPa):

136.73

Maximum Deformation (mm):

23.50

Plate Area (m2):

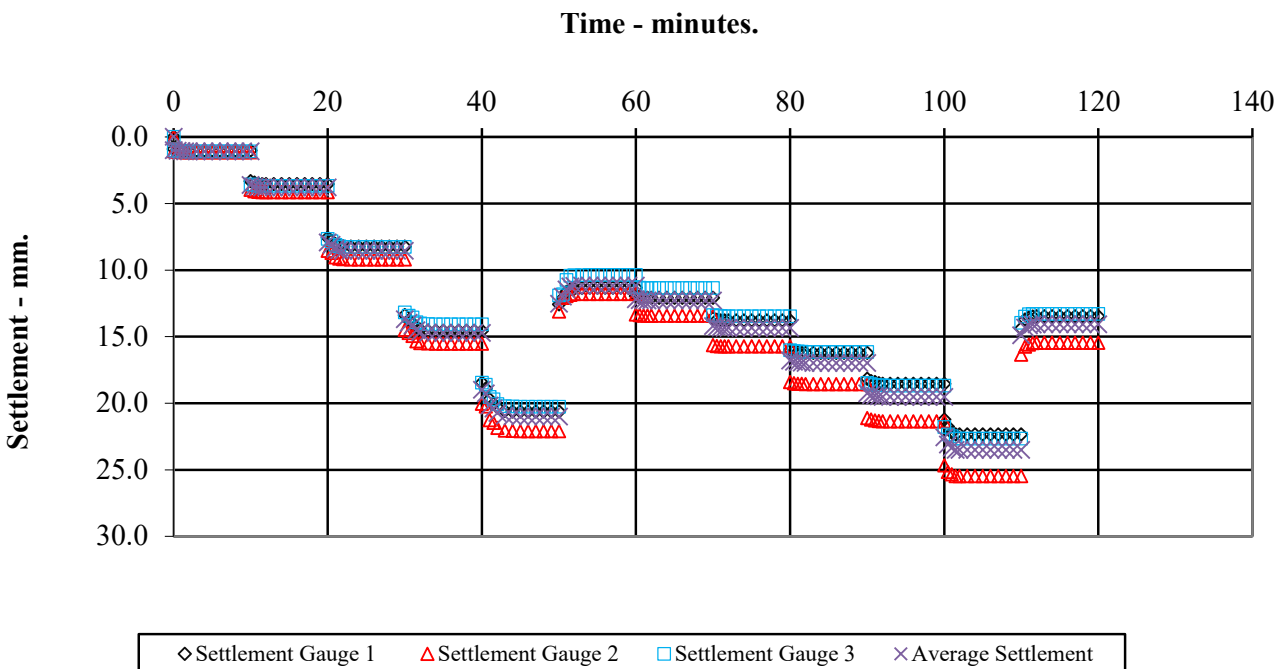
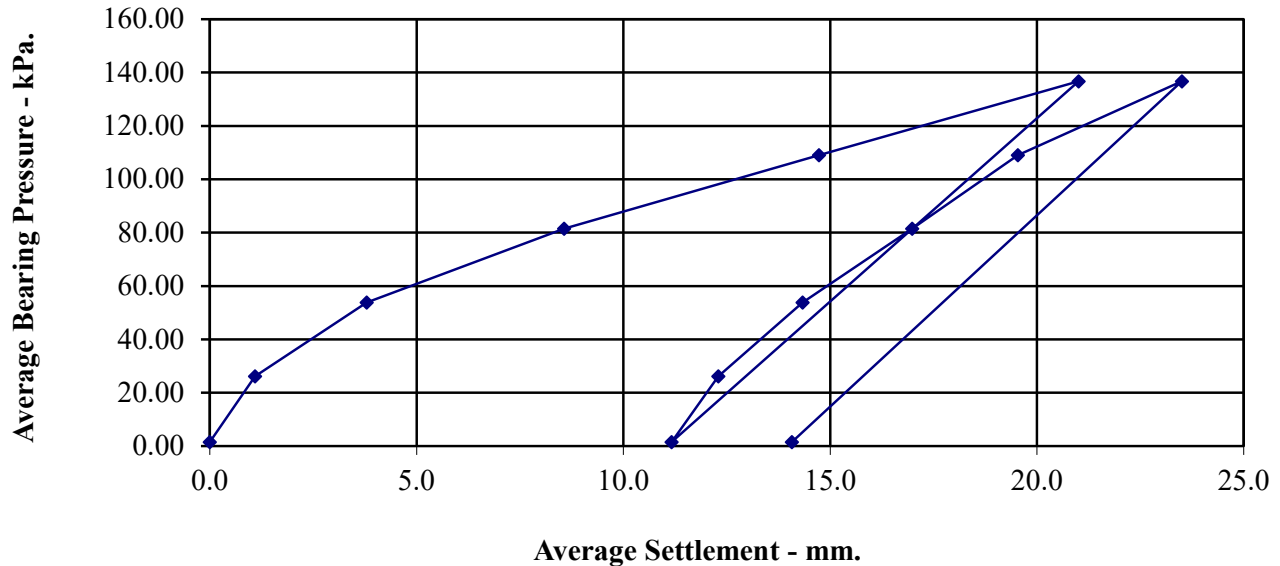
0.2922

Description:

Brown CLAY

Ratio of EV2/EV1

1.70



Calculation of Equivalent CBR Value from Plate Bearing Test

Design Manual for Roads and Bridges Volume 7: Pavement Design and Maintenance.

HD 25/94: Foundations

Date of Test	17-Apr-26
Test Ref	PBT 2
Depth (m)	0.45
Grid Ref	
Layer	
Conditions	Dry

Description	Brown CLAY
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Maximum Deflection	23.50	mm
Deflection required for CBR value	1.25	mm
Load at 1.25mm	28	kN/m²
Plate diameter	610	mm
Conversion factor for plate diameter	0.816	
 K₇₆₂(modulus of subgrade reaction) calculated using 1.25mm settlement	 18.1	 kN/m²/mm
 CBR Value	 1.5	 %



**Land off Tinker Lane, Rufforth,
York, YO23 3QA**

Contract No:
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VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test: 17-Apr-26

Test Ref: PBT 3

Grid Ref:

Depth (m): 0.50

Layer:

Conditions: Dry

Maximum Applied Pressure (kPa):

136.73

Maximum Deformation (mm):

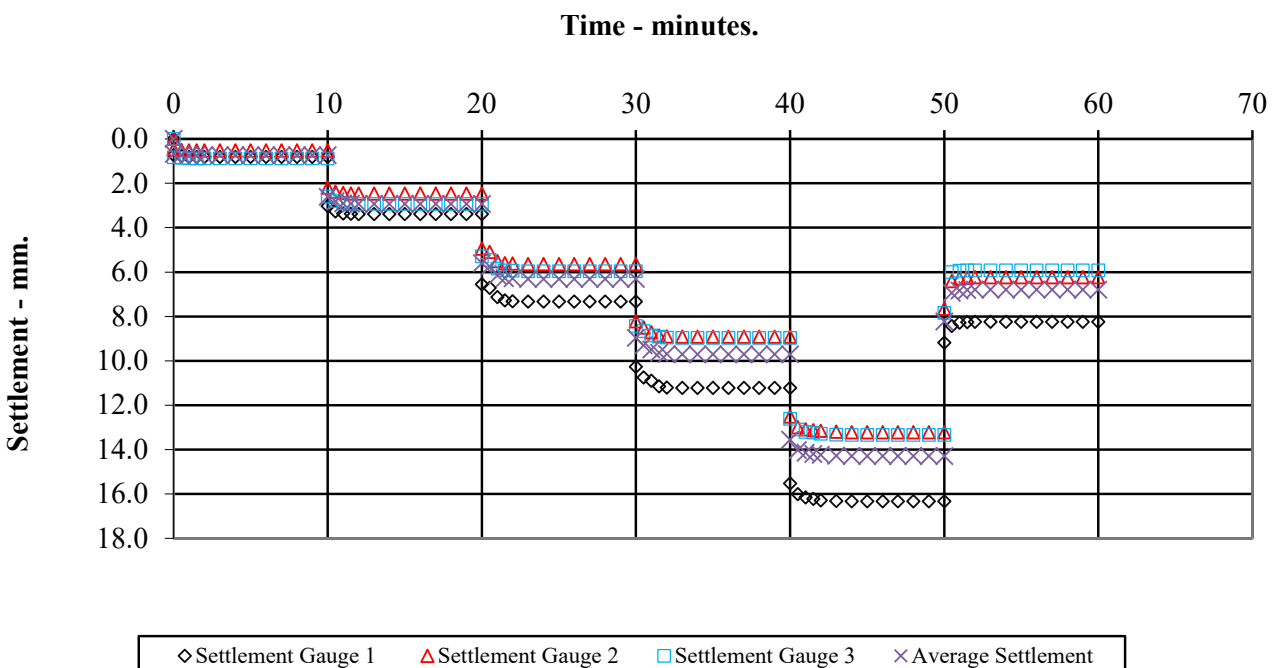
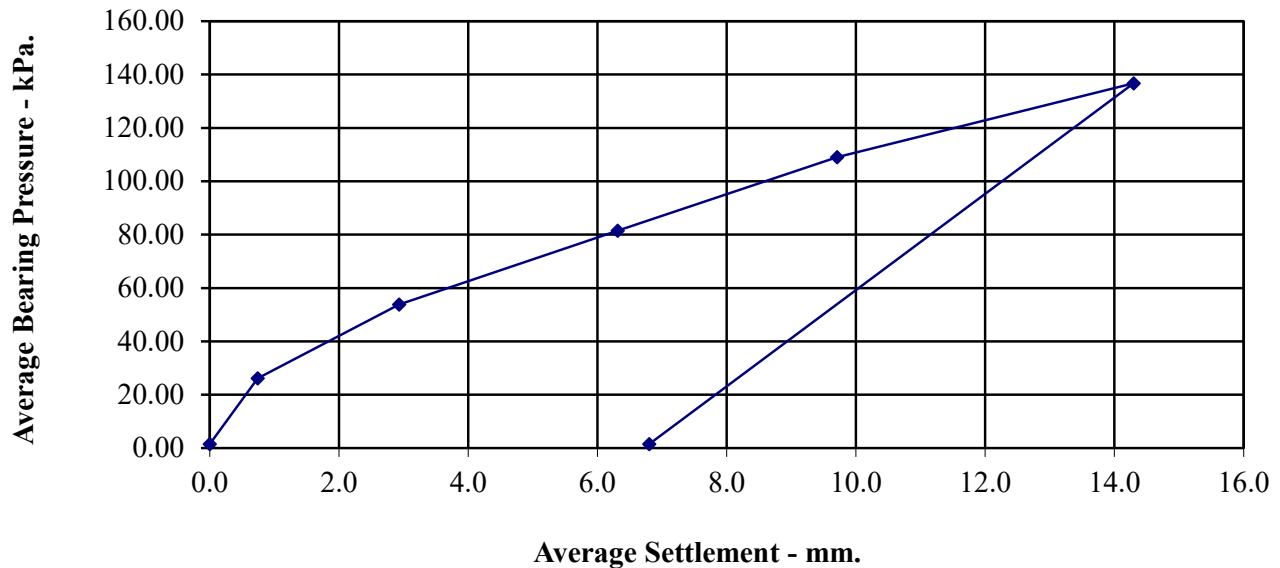
14.30

Plate Area (m²):

0.2922

Description:

Brown CLAY



Calculation of Equivalent CBR Value from Plate Bearing Test

Design Manual for Roads and Bridges Volume 7: Pavement Design and Maintenance.

HD 25/94: Foundations

Date of Test 17-Apr-26
Test Ref PBT 3
Depth (m) 0.50
Grid Ref
Layer
Conditions Dry

Description Brown CLAY

Maximum Deflection 14.30 mm
Deflection required for CBR value 1.25 mm
Load at 1.25mm 33 kN/m²
Plate diameter 610 mm
Conversion factor for plate diameter 0.816

**K₇₆₂(modulus of subgrade reaction)
calculated using 1.25mm settlement** 21.3 kN/m²/mm

CBR Value 1.9 %



Land off Tinker Lane, Rufforth, York

Contract No:
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VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test: 17-Apr-26

Test Ref: PBT 4

Grid Ref:

Depth (m): 0.50

Layer:

Conditions: Dry

Maximum Applied Pressure (kPa):

136.73

Maximum Deformation (mm):

19.62

Plate Area (m²):

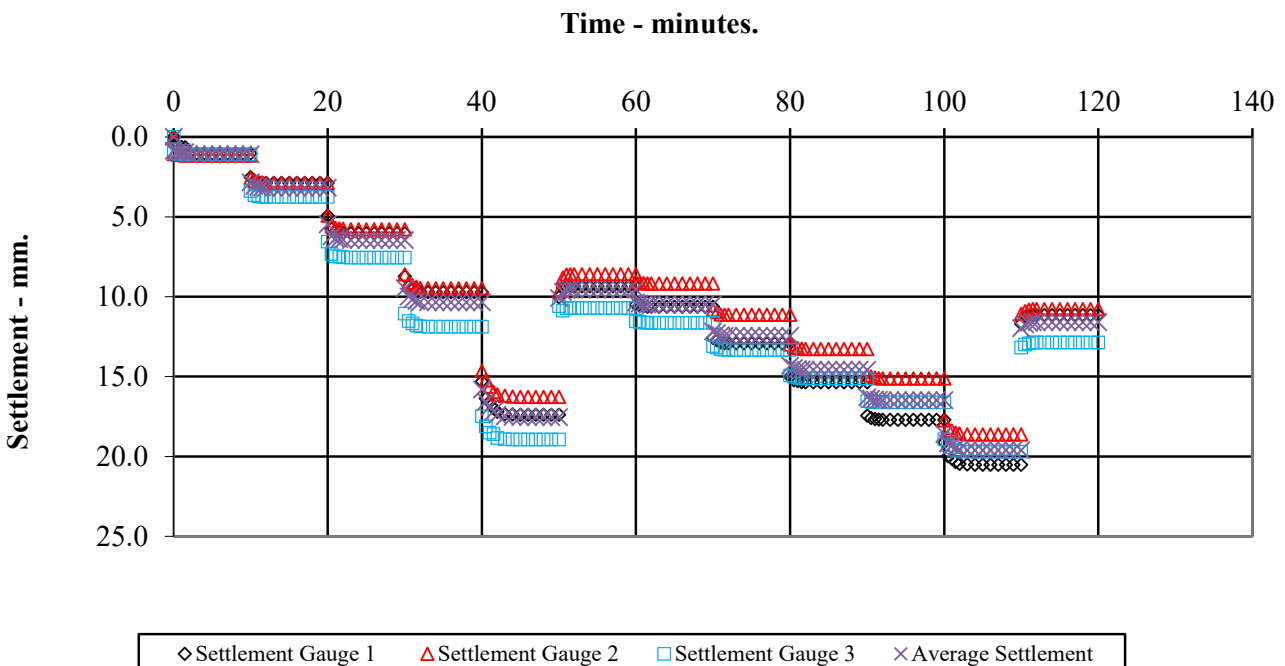
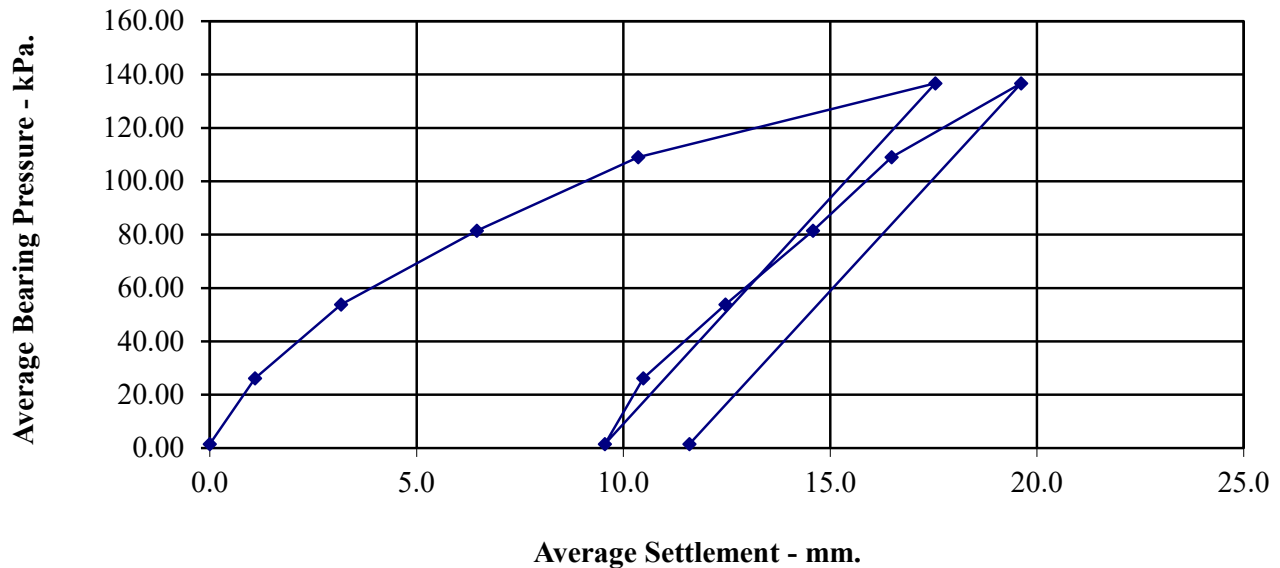
0.2922

Description:

Brown slightly gravelly CLAY

Ratio of EV2/EV1

1.74



Calculation of Equivalent CBR Value from Plate Bearing Test

Design Manual for Roads and Bridges Volume 7: Pavement Design and Maintenance.

HD 25/94: Foundations

Date of Test	17-Apr-26		
Test Ref	PBT 4		
Depth (m)	0.50		
Grid Ref			
Layer			
Conditions	Dry		
Description	Brown slightly gravelly CLAY		
Maximum Deflection	19.62	mm	
Deflection required for CBR value	1.25	mm	
Load at 1.25mm	28	kN/m ²	
Plate diameter	610	mm	
Conversion factor for plate diameter	0.816		
K ₇₆₂ (modulus of subgrade reaction) calculated using 1.25mm settlement	18.4	kN/m ² /mm	
CBR Value	1.5	%	

VERTICAL DEFORMATION TESTS.

BS 1377 : Part 9 : 1990.

Date of Test: 17-Apr-26

Test Ref: PBT 5

Grid Ref:

Depth (m): 0.40

Layer:

Conditions: Dry

Maximum Applied Pressure (kPa):

136.73

Maximum Deformation (mm):

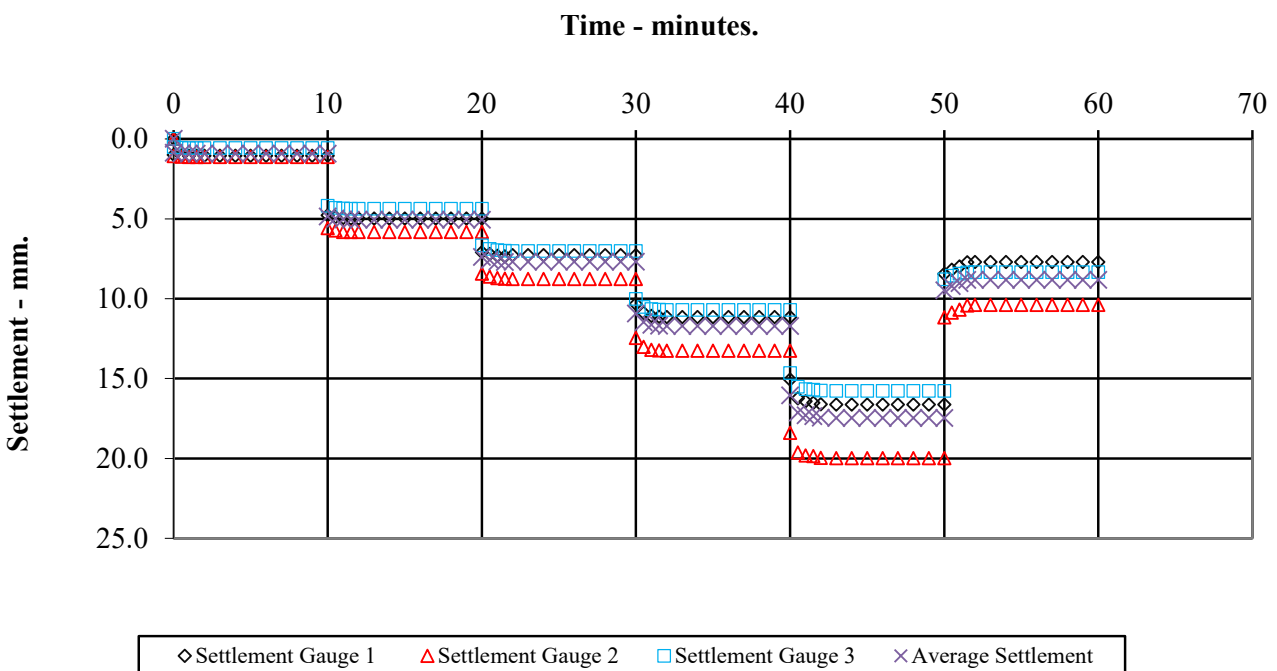
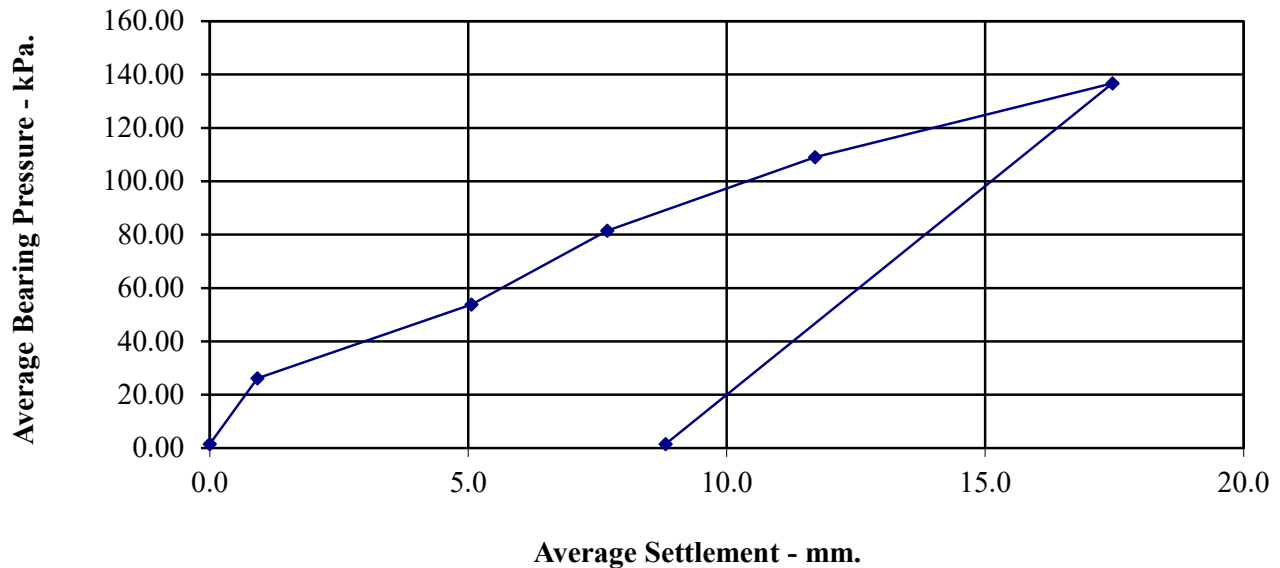
17.46

Plate Area (m²):

0.2922

Description:

Brown CLAY



Calculation of Equivalent CBR Value from Plate Bearing Test

Design Manual for Roads and Bridges Volume 7: Pavement Design and Maintenance.

HD 25/94: Foundations

Date of Test 17-Apr-26
Test Ref PBT 5
Depth (m) 0.40
Grid Ref
Layer
Conditions Dry

Description Brown CLAY

Maximum Deflection 17.46 mm
Deflection required for CBR value 1.25 mm
Load at 1.25mm 28 kN/m²
Plate diameter 610 mm
Conversion factor for plate diameter 0.816

**K₇₆₂(modulus of subgrade reaction)
calculated using 1.25mm settlement** 18.5 kN/m²/mm

CBR Value 1.5 %



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